

Lu Jiadong

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EDUCATION

National University of Singapore

Biomedical Informatics(Specialisation: Analytics)

Jilin University(Zhuhai)

Data Science and Big Data Technology

Singapore

Aug 2025 - Now

Zhuhai, China

Oct 2020 - Jul 2024

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java

Frameworks & Tools: PyTorch, Flask, PyQt5, Vue3, React, Git, Postman, Navicat, Docker, Jenkins, Overleaf, Trello

WORK EXPERIENCE

Software Development(Frontend) Engineer (7.2024 – 2.2024)

Jiusong Health Technology Co., Ltd.

- Responsible for front-end development of the Diabetes-AI-Agent Operations & Management Platform, building a visual monitoring and configuration center for AI service.
 - Developed a real-time chat dashboard via polling, enhancing control and observability over service processes.
 - Built a taskPackage calendar panel with TypeScript for view–logic separation, enabling batch operations, template import, and other features to streamline service cycle setup and management.
 - Implemented a SOP configuration tool using Vue Flow, enabling drag-and-drop orchestration of Agent-initiated and scheduled workflows, reducing operational complexity.
 - Tech stack:** Vue3, TypeScript, Pinia (with persistence), SCSS, Element-Plus; hooks/utilities; responsive design.
- Mini-program development: Developed the AI service activation module, including form validation, health assessment questionnaires. Leveraged SurveyJS to build logic-driven, style-rich questionnaires, resolving SurveyJS–Uniapp compatibility issues via WebView while ensuring smooth user experience.
- Backend & cloud deployment skills: Familiar with Node.js [GitHub](#) and RESTful APIs, experienced in designing scalable JSON structures and developing simple APIs with Spring Boot. Hands-on AWS experience, provisioning EC2/RDS, configuring VPC, and building CI/CD pipelines with Jenkins for front-end and back-end deployment.

LLM Agent Developer (1.2025)

Jiusong Health Technology Co., Ltd.

- Built a diabetes-AI-agent on the COZE platform: Used a decision-tree–like structure to analyze complex patient inquiries, accurately identifying multiple sub-intents, enabling autonomous retrieval strategy selection and dynamic API calling for retrieval and memory. Implemented multi-source result reranking to augment generation and stored key data in the patient database to support personalized interventions and scheduled task execution.
- Web scraping & AI knowledge base: Used Playwright to automate scraping of drug descriptions and food GI from websites with complex login and forms. Conducted data cleaning and structured ingestion to build an authoritative, AI-accessible knowledge repository.
- LLM SFT&RLHF in Diabetes Scenarios: Proficient in Transformer, RoPE, and Megatron pipeline/tensor parallelism; familiar with RL including PPO, DPO, and GRPO(with verl). Fine-tuned models for domain-specific tasks using LoRA and performed distributed full-parameter training on the base model. Built a diabetes multi-types instruction dataset(weight, diet, lifestyle) and fine-tuned models how to extract key info in diabetic context for the Agent memory module. Deployed with vLLM, boosting info extraction performance in diabetes scenarios. [HuggingFace](#)

PROJECTS

- Typhoon Track Data Analysis and Forecasting**, Processed CMA (1949–2018) typhoon data in Python, extracting features such as pressure and wind speed. Built interactive visualizations with ECharts to display annual counts and average wind intensity. [GitHub](#) Combined CMA and ERA5 multi-source data to develop a CNN–Transformer spatiotemporal encoder–decoder and applied ensemble learning, outperforming the baseline in 24-hour trajectory forecasting. [Publication](#) Developed an interactive typhoon track visualization system using PyQt5. [GitHub](#)
- WeChat Message Data Analysis**, Parsed encrypted WeChat databases on MacOS. Analyzed data in Python and visualized daily message trends, message keywords, and message types using ECharts. [GitHub](#) [55 stars](#)

ADDITIONAL INFORMATION

- English Reading & Writing Skills**, IELTS: Reading 7.0 / Writing 6.5; capable of rapidly learning new technologies by reading official documentation and experienced in writing technical documents.
- Agile (Scrum) Fundamentals**, trained in sprint planning, daily stand-ups, rapid review, and retrospectives.
- Full-time Vacation & Second-Semester Internship**, credited as capstone project.